Hsun-Yu Kuo

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Research Interest

- **Machine Learning**
  Data-Centric AI; Reinforcement Learning; Extrapolation; Robustness; Causality; AI for Social Good

Education

- **Swiss Federal Institute of Technology (EPFL)** — **Lausanne, Switzerland**
  M.Sc. in Computer Science (Specialization in Computer Science Theory) — *Sept. 2024 – Present*
  - **Coursework**: Machine Learning; Advanced Algorithms; Causal Thinking; Mathematics of Data: From Theory to Computation; Markov Chains and Algorithmic Applications; Deep Learning; Reinforcement Learning; Modern Natural Language Processing
  - Completed mandatory military service (Mar. 2025 – Jun. 2025)
- **National Taiwan University** — **Taipei, Taiwan**
  M.Sc. in Data Science — *Feb. 2023 – June 2024*
  - Paused studies after June 2024 to pursue a master's degree at EPFL
  - **Coursework**: Machine Learning; Natural Language Processing; Statistical Foundations of Data Science (I); Computational Methods for Data Science; Big Data Systems
- **National Taiwan University of Science and Technology** — **Taipei, Taiwan**
  B.Sc. in Applied Science and Technology, Major in Computer Science — *Sept. 2018 – June 2022*
  - Graduated 1st in the program (2022)
  - **Coursework**: Statistics (I); Probability and Statistics; Discrete Mathematics; Linear Algebra; Calculus (I); Algorithms; Data Structures; Database Systems; Operating Systems; Digital Logic Design; Computer Networks; Webpage Development; Object-Oriented Programming; Introduction to Computers; Value of AI and Data; Introduction to Fuzzy Systems; Machine Learning Foundations

Research and Industry Experience

- **MLO Lab, EPFL** — **Lausanne, Switzerland**
  Semester Project Student — *Sept. 2025 – Present*
  - **Advisors**: Martin Jaggi; El Mahdi Chayti
  - Exploring the Halting Problem in recurrent structures (looped transformers).
- **Robust Machine Learning Group, Max Planck Institute for Intelligent Systems** — **Tübingen, Germany**
  Research Intern — *June 2025 – Aug. 2025*
  - **Advisors**: Wieland Brendel; Patrik Reizinger
  - Exploring out-of-distribution generalisation and extrapolation.
- **CKIP Lab, Academia Sinica & IR Lab, National Taiwan University** — **Taipei, Taiwan**
  Research Assistant — *Feb. 2023 – June 2024*
  - **Advisors**: Wei-Yun Ma; Pu-Jen Cheng
  - Data weighting for synthetic data to better match real-data distributions.
- **Industrial Technology Research Institute** — **Taipei, Taiwan**
  Intern — *July 2021 – Sept. 2021*
  - Engineered short- and long-term load forecasting methods for Taiwan Power Company.
- **Wireless System Lab, National Taiwan University of Science and Technology** — **Taipei, Taiwan**
  Research Assistant — *Sept. 2019 – Jan. 2021*
  - **Advisor**: Chin-Ya Huang
  - Devised an MTC scheme for dynamic CBRS channel allocation.
  - Implemented RLNC and Galois field arithmetic in P4; simulated with ONOS/Mininet.

Papers

[P1] **Hsun-Yu Kuo**, El Mahdi Chayti, Patrik Reizinger, Martin Jaggi & Wieland Brendel, "Stabilizing Extrapolation in Looped Transformers via Learned Stochastic Stopping". *Under Review*

[P2] **Hsun-Yu Kuo**, Yin-Hsiang Liao, Yu-Chieh Chao, Wei-Yun Ma & Pu-Jen Cheng, "Not All LLM-Generated Data Are Equal: Rethinking Data Weighting in Text Classification". In *Proceedings of the 13th International Conference on Learning Representations* (ICLR 2025) (**Spotlight**, top 5.1%)

[P3] **Hsun-Yu Kuo**, Szu-Yu Liu, Chin-Ya Huang, Yu-Chi Chen & Meng-Hua Xie, "Reliable Data Transmission through Private CBRS Networks". In *Preprint* (2023)

[P4] **Hsun-Yu Kuo**, T.-W. Liu, Y.-P. Huang, *et al.*, "Differential Diagnostic Value of Machine Learning–Based Models for Embolic Stroke". *Clinical and Applied Thrombosis/Hemostasis* (2023)

[P5] Hsuan-Min Wang, Yo-Ping Huang, **Hsun-Yu Kuo**, *et al.*, "The modified spatial context memory test for assessing cognitive aging in middle-aged and older adults". *Neuroscience Research* (2019)

Teaching Experience

- **Teaching Assistant, National Taiwan University** — **Taipei, Taiwan**
  System Programming — *Sept. 2023 – Jan. 2024*
  - Led TA sessions and supported 100+ students.
  - Designed a context-switch simulation using non-local jumps and signals for a course assignment.

Honors and Awards

- **Phi Tau Phi Scholastic Honor Society of the Republic of China** — **Taiwan**
  Honorary Membership — *2022*
- **Mengjia Longshan Temple Scholarship** — **Taiwan**
  Academic Excellence Scholarship (Scholarship for Excellent Academic Performance) — *2021*
- **Academic Excellence Award** — **Taiwan**
  Seven consecutive awards for excellent academic performance — *2018–2022*
- **Kanagawa International Science Forum** — **Kanagawa, Japan**
  1st Place (Outstanding Poster Presentation Award) – Engineering Category — *2017*
- **National Skills Competition** — **Taiwan**
  1st Place (Golden Award) – Software Applications Development — *2016*

Professional Service

- **Conference Reviewer**
  ICLR 2026, NeurIPS 2026
- **Volunteer**
  SIGIR 2023